

# Zebra TC57 Hardware Repair 002

## Documentation

### Device Information

**Device Type:** Zebra TC57 Mobile Computer

**Department/Location:** Enterprise Operations

**Repair Type:** Hardware Repair / Troubleshooting - Display Assembly Replacement & Advanced Support Escalation

**Technician:** Ridley Lherisse

**Date Received:** 09/26/2026

**Date Completed:** Pending Escalation Resolution

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### Issue Reported

#### Problem Description:

The Zebra TC57 device was received with a damaged display assembly. The device powered on and scanner functionality remained operational; however, the display showed a grey screen with no visible graphics output.

#### Initial Symptoms:

- Display was damaged/unusable
  - Device functionality needed to be evaluated
  - Hardware inspection required to determine whether the issue was limited to the display assembly or involved internal components
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# Troubleshooting Process

## Initial Assessment

1. Confirmed the device condition before beginning repair.
  2. Powered down the device and disconnected power source.
  3. Inspected the exterior housing for additional damage.
  4. Evaluated whether the issue appeared related to:
    - LCD display assembly
    - Display connector/flex cable
    - Internal motherboard components
    - Other hardware failures
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## Repair Procedure

### Device Disassembly

1. Powered down & removed battery of the Zebra TC57 and prepared the device for hardware service.
2. Applied 91% isopropyl alcohol in controlled amount around the edges of the display screen to help loosen the adhesive bond and safely separate the screen from the device housing
3. Carefully separated the device display screen from the housing using appropriate electronics repair tools while avoiding unnecessary pressure on internal components, connectors, and cables.
4. Removed the damaged display of assembly and inspected the internal housing area of any signs of
  - Physical Damage
  - Loose connections
  - Damaged flex cables
  - Debris or contamination
  - Potential damage to motherboard components or chips
5. Before installing the replacement display, I inspected the housing, motherboard area, connectors, and surrounding components to confirm the issue was isolated to the display assembly and that no additional damage was present.
6. Cleaned the repair area by removing broken glass fragments and leftover adhesive residue. Used controlled cleaning methods such as a cotton swab with the 91% iso alcohol, helpful with removing dust, fingerprints, oils, glass fragments & residue.

7. Installed the replacement LCD/display assembly, ensuring proper alignment and secure connection of all components.
  8. Reconnected the battery, wait 30 seconds to a minute. Powered on the zebra device. Tested the touchscreen, brightness and display connections.
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## Internal Inspection

Performed inspection of internal components before installing replacement parts.

Inspected:

- Motherboard condition
- Display connector
- Flex cable connection
- Internal chips/components
- Signs of physical damage or debris

**Inspection Result:**

**- Motherboard LED backlight component was found missing during inspection.**

**- Display failure was determined to involve a motherboard-level backlight circuit issue.**

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## Replacement Installation

1. Installed replacement LCD/display assembly.
  2. Ensured the display assembly was properly aligned and seated.
  3. Verified internal connections were secure.
  4. Reconnected battery after completing installation.
  5. Allowed device components to stabilize before powering on.
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# Testing and Verification

- ✓ Device powered on successfully
- ✓ Scanner functionality remained operational
- ✓ Replacement display assembly connection was verified
- ✓ Display connector and flex cable connections were inspected
- ✓ Internal components were inspected for additional damage

## FINDINGS:

- ⚠ Full display functionality was not restored
  - ⚠ A motherboard-level LED backlight circuit component issue was identified during troubleshooting. (missing)
  - ⚠ The device was documented and escalated to advanced hardware support for further board-level repair.
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# Final Result

## Repair Status: Escalated to Advanced Hardware Support

The Zebra TC57 display assembly was replaced and initial troubleshooting was performed. During internal inspection, a motherboard-level backlight circuit issue was identified involving a missing/damaged LED backlight component. Due to the requirement for board-level component repair, the device was escalated to experienced hardware technicians with the capability to perform motherboard-level repair. All troubleshooting steps and findings were documented to support the next stage of repair.

# Lessons Learned / Notes

- Hardware issues should be diagnosed before replacing components.
- Proper inspection of connectors and internal components helps prevent repeat failures.
- Verification testing after repair is essential before returning equipment to operation.